# Wine House

## Smart Wine Restaurant Management System

Runs locally · offline-resilient · role-based

Version 0.1.0 · Introduction · 2026

# At a glance

Measured, tested, working software

| | | | |
|---|---|---|---|
| **16**<br>Django apps | **129**<br>database tables | **19**<br>user roles | **13**<br>built-in reports |
| **380**<br>automated tests · all passing | **73%**<br>line coverage (all code) | **0**<br>known runtime vulnerabilities | **29**<br>runtime dependencies |

# Why is a wine restaurant different?

What generic POS software cannot solve

- Wine sells in two units— bottle and glass — but stock is held in one.
- An opened bottle starts to deteriorate; when it was opened and how it is preserved matters.
- The gap between expected and actualglasses per bottle is direct lost profit.
- Wine has a drink window; a bottle not sold in time loses value.
- A few degrees of cellar temperature drift can damage thousands in inventory.
- A sommelier's recommendation must be limited to what is actually in stock.

# What does Wine House do?

One application, one database

- **Floor operations** — table map, reservations, orders, kitchen display, payment, day close
- **Digital wine cellar** — identity, layout, lot tracking, by-the-glass service, yield analysis
- **Menu and costing** — recipes, portion cost, allergens, dynamic pricing
- **Inventory and purchasing** — FIFO/FEFO, expiry, waste, suppliers, quotation comparison
- **Customers and staff** — GDPR-compliant CRM, loyalty, shifts, performance
- **Reporting** — 13 reports, PDF / Excel / CSV
- **Artificial intelligence** — sommelier assistant, provider-agnostic, privacy-preserving

# Design principles

Every decision has a rationale

| Principle | What it means |
| --- | --- |
| Local-first | The restaurant keeps working when the internet drops |
| AI is optional | All core functions work fully with AI disabled |
| Financial records are frozen | Changing a menu price never alters a past bill |
| Trace instead of delete | Corrections use reversing entries; the audit trail survives |
| Safe by default | Dangerous features ship switched off |
| Least privilege | Each user sees only their own job |

# Floor Operations

From table to till

# Table map

The whole floor at a glance

- Section-based layout: main room, terrace, bar, cellar room
- Eight table states coded by colour and text— readable in low light
- Open order number and current total shown on the table
- Click a table to jump straight to its order
- 44-pixel touch targets for POS terminals

# Reservation conflict detection

The same table cannot be double-booked

1. Create reservation
2. Assign table
3. Do the time windows overlap?
4. Is capacity sufficient?
5. Accept or reject

On conflict the system refuses the assignment and names the clashing reservation.

# Orders: why freeze the price?

The foundation of financial accuracy

## If it were not frozen

· The menu price changes today

· Yesterday's bill total changes too

· Accounts no longer reconcile with the till

· A historical report returns different numbers each run

## In Wine House

· Price, tax, cost and item name are copied at line creation

· Later menu changes never touch a past bill

· A report run a year later returns the same result

· Day-close totals are frozen a second time

# Kitchen order tickets and displays

KOT generation and KDS flow

- Sending an order produces one ticket per station
- Food to the kitchen, drinks to the bar, wine to the sommelier — automatically
- Ticket colour follows elapsed time: grey → amber → red (overdue)
- Allergen warnings appear as a red line on the ticket
- Ticket flow: Queued → Preparing → Ready → Served
- A sent line is never sent twice

# Bill splitting and payment

Seat-based splitting, multi-method payment

- Automatic split by seat number — each seat becomes its own order
- Order merging: when two tables join, so do their bills
- Cash, card, meal card, transfer, voucher, loyalty points
- Partial payments supported; the outstanding balance is always visible
- Card commission is calculated automatically and reported net
- Sandbox mode:no real money moves in this release

# Day close and cash reconciliation

Closed once, then frozen

- The day cannot closewhile orders are open — no missing bill slips through
- Expected cash is compared with counted cash and the variance shown
- Payment-method breakdown, voids and refunds are computed
- All totals are frozeninto a daily snapshot
- Reports read from that snapshot — history never shifts

# Digital Wine Cellar

The distinguishing core

# Wine identity

Everything a sommelier wants to know

- Producer, vineyard, country → region → appellation hierarchy
- Grape composition with percentages — 60% Öküzgözü, 40% Boğazkere
- Vintage, alcohol, bottle size, sulphite and vegan information
- Tasting profile: body · acidity · tannin · sweetness (0–5 scale)
- Aroma notes, serving temperature, decanting time, glass type
- Drink window: from · peak · until
- Barcode, internal QR code, label photograph

# Cellar layout and storage conditions

The physical reality of inventory

- Hierarchical layout: cellar → cabinet → rack → bin → display
- Target temperature, humidity and tolerance per location
- Readings are recorded; out-of-range drift raises an alert
- Occupancy rate and bottles stored are shown live
- Private-cellar clients' bottles are kept separate and excluded from house valuation

# Lot tracking and FEFO

Which bottle should leave first?

- Every purchase lot is tracked separately: receipt date, unit cost, supplier, location
- FEFO— the lot that must be consumed first is opened first
- Lot-level cost is the basis of accurate profitability
- Invoice reference and supplier linkage are preserved
- Remaining bottles can never exceed received bottles (database constraint)

# How a glass leaves stock

The system's finest business rule

1. Pour a glass (150 ml)
2. Is there an open bottle with enough volume?
3. If not: pick lot by FEFO → deduct 1 bottle from stock
4. Reduce the open bottle by 150 ml
5. Write the pour record (sale / tasting / comp / spillage)

When less than half a glass remains, the bottle is marked finished automatically. Two waiters cannot open the last bottle simultaneously — row locking prevents it.

# Yield analysis

750 ml bottle · 150 ml pour · 5 glasses in theory

| | | |
|---|---|---|
| 5<br>expected glasses | 4<br>actual glasses | 600 ml<br>served |
| 80%<br>realised yield | 150 ml<br>lost | 1 glass<br>missed revenue · per bottle |

The gap between theoretical and actual points to spillage or over-pouring. The glass-vs-bottle report shows this bottle by bottle.

# Open bottles and fault tracking

Where is the loss occurring?

- Freshness window by method: standard 48 h · preservation 1 week · needle system 30 days
- Bottles past their window are flagged on screen
- Fault records: cork taint (TCA), oxidation, reduction, heat damage, spillage, breakage
- Each fault deducts stock and computes the estimated loss
- Resolution is tracked: supplier claim · written off · replaced for the guest

# Drink window and cellar valuation

Inventory is an asset that changes over time

- Each wine gets a status: too young · ready · at peak· drink soon · past peak
- Bottles approaching the end of their window move to a priority list
- Cellar valuation: cost value, retail value, potential margin
- Remaining volume in open bottles counts towards valuation
- Wines below their minimum bottle count raise a critical-stock alert

# Duplicate and anomaly detection

Data quality does not maintain itself

- The same barcode used on two different records
- Same producer + very similar name + same vintage (fuzzy match)
- Implausible vintage — a future year or more than 120 years old
- Price anomaly — selling price below purchase price
- The system never deletes or merges anything on its own— it only raises a review flag

# Menu, Inventory and Costing

The arithmetic of profit

# Recipes and portion costing

The real cost of every plate

- Recipe lines link to stock items
- Portion cost is derived from the weighted averageunit cost of stock
- A waste allowance percentage is added to the cost
- Gross margin, margin percentage and food-cost ratioare automatic
- Allergen and dietary tags (vegetarian, vegan, gluten-free) attach to items
- Dynamic pricing: happy hour, campaigns, windows that cross midnight

# Menu engineering

What stays, what changes?

| Quadrant | Popularity | Margin | Action |
|---|---|---|---|
| Star | High | High | Protect and promote |
| Plowhorse | High | Low | Reduce cost or revisit price |
| Puzzle | Low | High | Promote; add to sommelier picks |
| Dog | Low | Low | Consider removing from the menu |

Kasavana-Smith method · computed automatically from sales data

# Inventory, purchasing and waste

Full traceability from receipt to issue

- FIFO or FEFO — chosen per item
- Alerts for lots approaching their expiry date
- Goods receipt: partial rejection, cold-chain check, automatic lot creation
- Stock counts: system versus counted, variance posted as a movement
- Waste records with 11 reasons: spoilage, expiry, breakage, spillage, staff meals…
- Reorder suggestions from minimum levels — the system suggests, never orders
- Quotation comparison: cheapest and fastest flagged separately

# Stock movements are immutable

The balance is always verifiable

- Every receipt and issue creates its own movement record
- Once written it cannot be updated or deleted— enforced in code
- Corrections are made with a reversing movement
- Stock balance can always be reconciled against the sum of movements
- The same immutability applies to loyalty points and the audit log

# Customers, Staff and Reports

# GDPR-compliant customer management

Data protection is not an afterthought

- Phone, email and identity data are stored encrypted
- Search over encrypted fields uses an irreversible HMAC digest
- Separate consent per purpose: service, email, SMS, profiling, AI processing
- Withdrawing consent does not delete the record — a withdrawal timestamp is written for proof
- Data portability: a customer's entire record exports in one click
- Right to erasure: personal data is anonymised, financial records retained by law
- Birth yearis not stored — only day and month (data minimisation)

# 13 reports · 3 formats

With guaranteed Turkish character rendering

- Sales: daily summary · products and categories · payment mix · forecast
- Wine: profitability · glass vs bottle · cellar valuation
- Inventory: waste analysis · stock valuation
- Operations: table turnover · staff performance · menu engineering
- AI: cost and token usage
- PDF · Excel · CSV— ç ğ ı İ ö ş ü render correctly in all three

# How Turkish characters are guaranteed

Where most systems break silently

| Format | The problem | Wine House solution |
| --- | --- | --- |
| CSV | Excel opens the file with the wrong encoding | Written with a UTF-8 BOM; survives a double-click |
| Excel | Numbers written as text | Numbers stay numeric; locale-aware number and date formats |
| PDF | Built-in fonts do not cover Turkish | A system TrueType font is located and embedded |
| PDF (no font) | Silently corrupt output | An explicit error is raised — no broken PDF is produced |

# Artificial Intelligence

An assistant, not an authority

# Provider-agnostic architecture

Never locked to a single vendor

- LM Studio(local) · Anthropic Claude· NVIDIA· Mock(offline testing)
- Model names are never hard-coded — always read livefrom the provider
- Task-aware model selection: chat, coding, vision, analytical, embedding
- Timeouts, retries, exponential backoff, request cancellation
- Fallback chain: if one provider fails, the next is tried
- Mock always sits last— the AI flow never brings the app down

# Privacy: what reaches the cloud?

The user always sees it

## Default behaviour

· The default provider is local LM Studio
· Customer data never leaves the premises
· `Local only` mode disables the cloud entirely
· Exceeding the monthly cost cap blocks cloud calls

## If a cloud call is needed

· Email, phone, national ID, IBAN, card number are masked
· API keys are never transmitted under any circumstance
· The user sees exactly what was sent on screen
· A summary reports how many items of each type were masked

# AI cannot write to the database

A hard boundary

- Every suggestion is persisted as a recordand awaits human approval
- No price, stock, menu item or pairing changes without that approval
- Sommelier suggestions are drawn only from wines currently in stock
- The system prompt explicitly forbids health claims and medical advice
- The 18+ limit and responsible consumption appear on every suggestion screen
- Every call is logged: provider, model, tokens, cost, latency, outcome

# Security and Quality

# 19 roles · least privilege

Everyone sees only their own job

| Role | Permissions | Cannot do |
|---|---|---|
| Waiter | 18 | Cellar management · user creation · day close |
| Sommelier | 41 | Day close · taking payment · staff management |
| Cashier | 12 | Menu changes · stock adjustments |
| Auditor | 101 | No write operations at all |
| System administrator | 519 | — |

# Second approval and audit log

Separation of duties

- Critical operations need approval: voids, refunds, over-limit discounts, comps, stock adjustments, restores
- A user cannot approve their own request
- Without approval the operation is not applied — a request is raised and the process stops
- The audit log is immutable and undeletable— enforced in code
- Permission denials are logged too; unauthorised attempts are traceable
- Records past their retention period are archived before removal

# Backup and restore

A backup is not a backup until a restore is proven

- Encrypted ZIP archive + SHA-256 integrity digest
- Every backup is verified automaticallyafter creation
- Restores require second approval
- An automatic safety backupis taken before writing to the live database
- The default target is the test database — drills are possible
- The retention policy always keeps at least three backups
- Daily automated backup via Windows Task Scheduler

# Quality indicators

Measured and reported

**380**
tests · all passing

**73%**
line coverage (all code)

**0**
known runtime dependency vulnerabilities

**0**
high/medium static-analysis findings

**0**
secret-scan findings

**23**
database migrations

All figures are measured from the repository at release time: pytest, coverage, pip-audit/osv-scanner (runtime dependencies), bandit and semgrep (medium+), gitleaks. Details: PUBLIC_RELEASE_MANIFEST.json

# Technical profile

| Layer | Choice |
| --- | --- |
| Language / framework | Python 3.11+ · Django 5.2 · Django REST Framework |
| Database | SQLite (WAL) · optional PostgreSQL |
| Interface | Bootstrap 5.3 + HTMX — bundled locally, no CDN |
| Encryption | Fernet (AES-128-CBC + HMAC-SHA256) |
| Reporting | ReportLab (PDF) · openpyxl (Excel) |
| Artificial intelligence | Custom httpx-based adapter layer |
| Installation | A single PowerShell script · double-clickable .bat files |

# Platforms & Download

Public GitHub repository and ready-made packages

- Windows 10/11 (x64)& macOS (Apple Silicon)— Download: GitHub Releases (v0.1.0)
- PublicGitHub repository: github.com/Azizsekerdil/smart-wine-restaurant-management-system
- Release v0.1.0 ships two packages: Windows ZIP+ macOS ZIP

# An honest status report

No unfinished feature is presented as finished

## Ready and working

· Floor operations · KDS · payment · day close
· Wine cellar · by-the-glass service · yield analysis
· Inventory · purchasing · waste · stock counts
· CRM · GDPR · staff · 13 reports
· AI layer · backups · 19 roles

## Experimental or planned

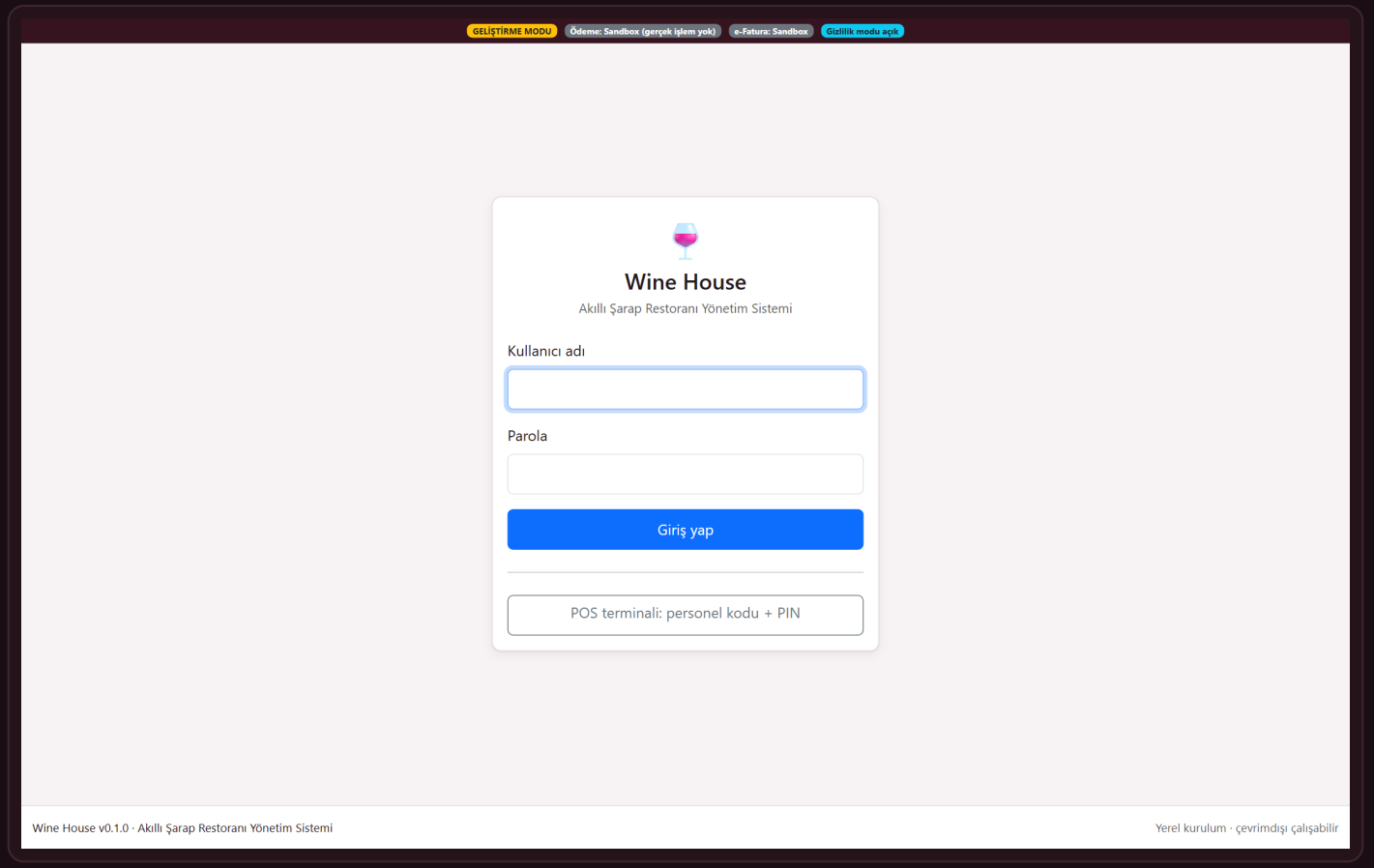
· Live payment and e-invoicing (adapter only)
· Thermal printer driver
· Bulk email/SMS dispatch
· Cellar sensor integration
· Label image analysis · multi-terminal sync

Every feature carries a badge in the application: Ready · Experimental · Planned

# Screens from the Application

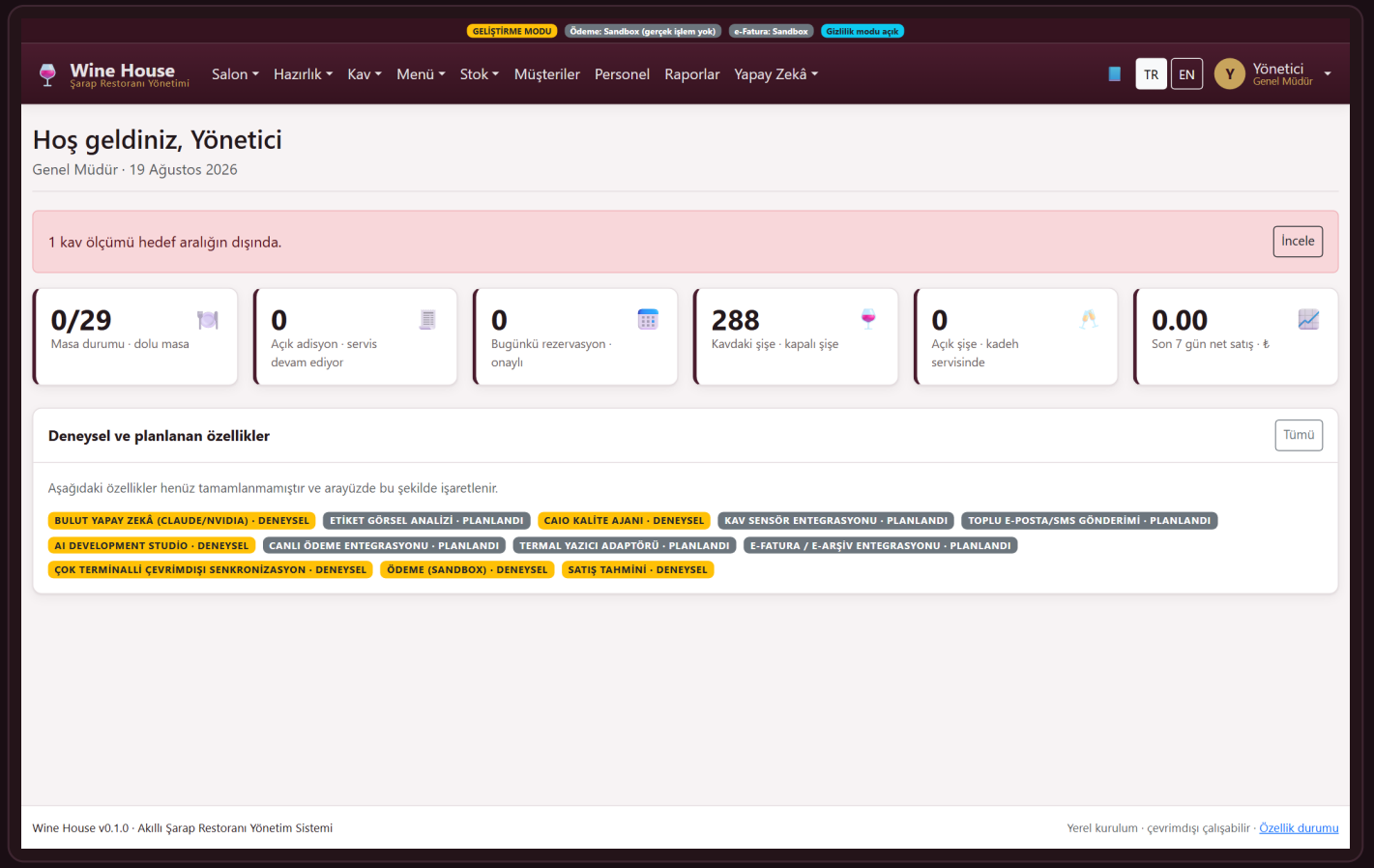
Real interface · synthetic demo data · v0.1.0 · UI shown in Turkish (EN toggle built in)

# Sign-in screen



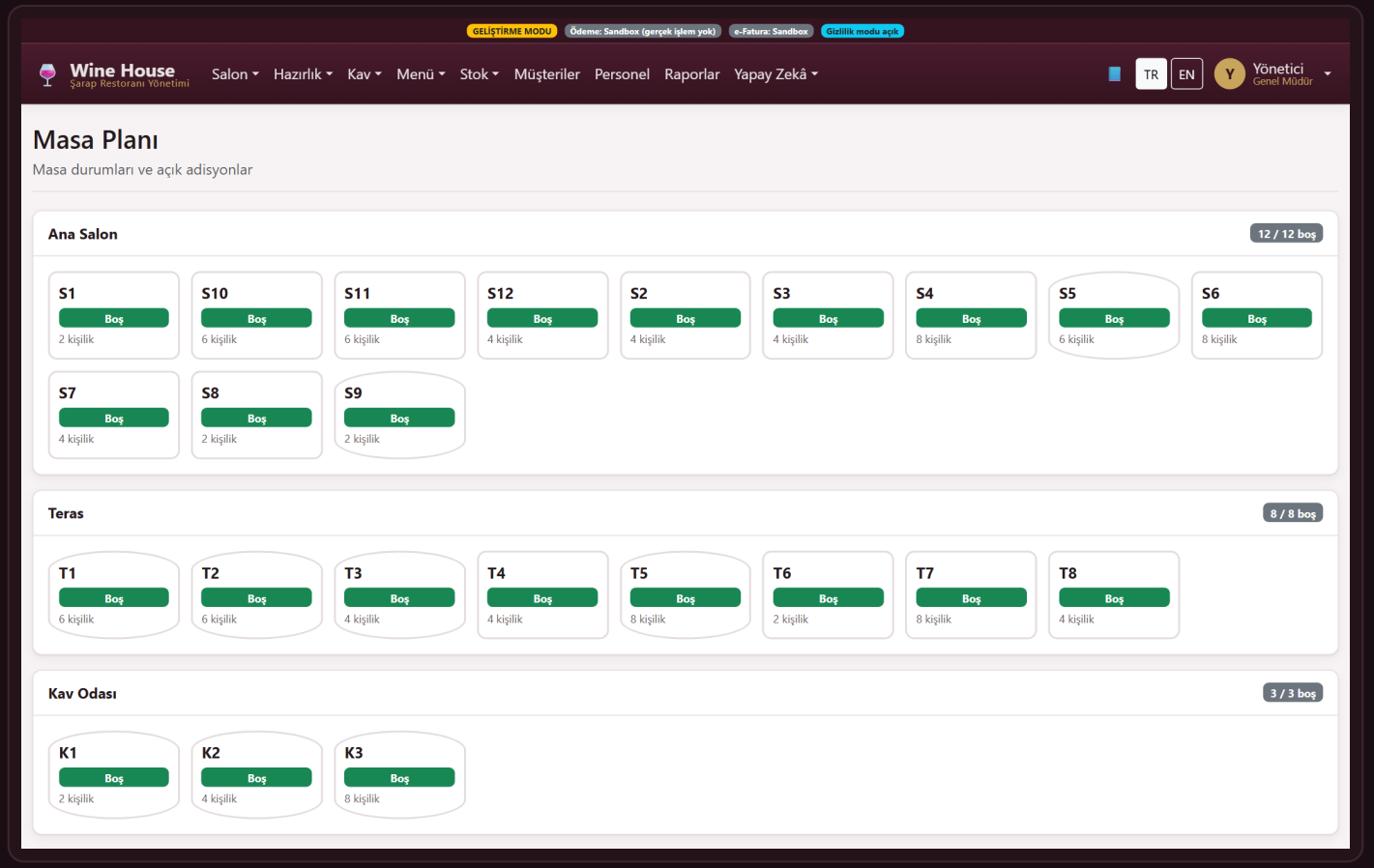

Password or quick PIN sign-in · role-based access starts with 19 roles

# Main dashboard



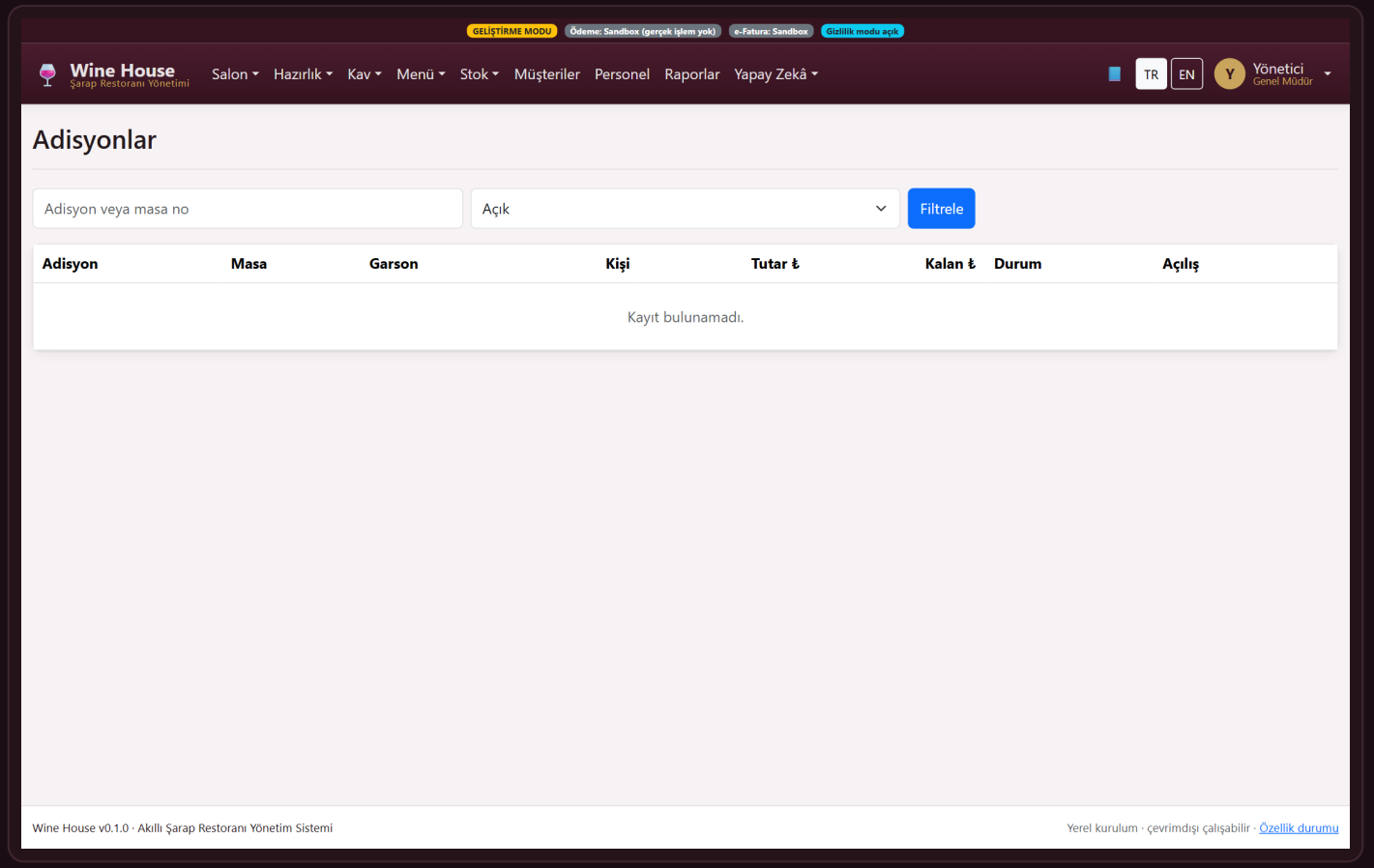

Tables, open orders, reservations, cellar and sales at a glance; experimental features are clearly badged

# Table map



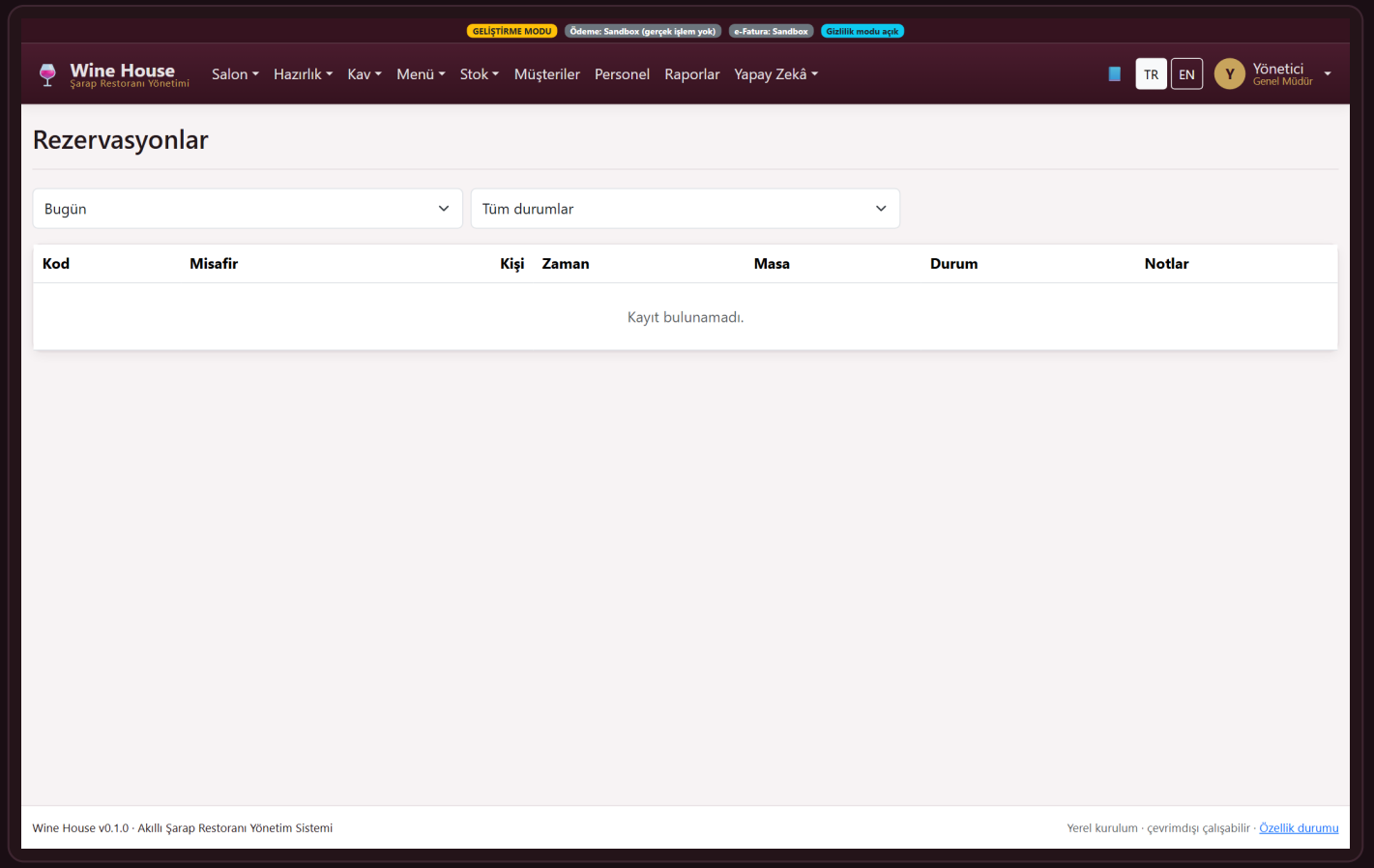

Live table status by section with links to open orders

# Orders



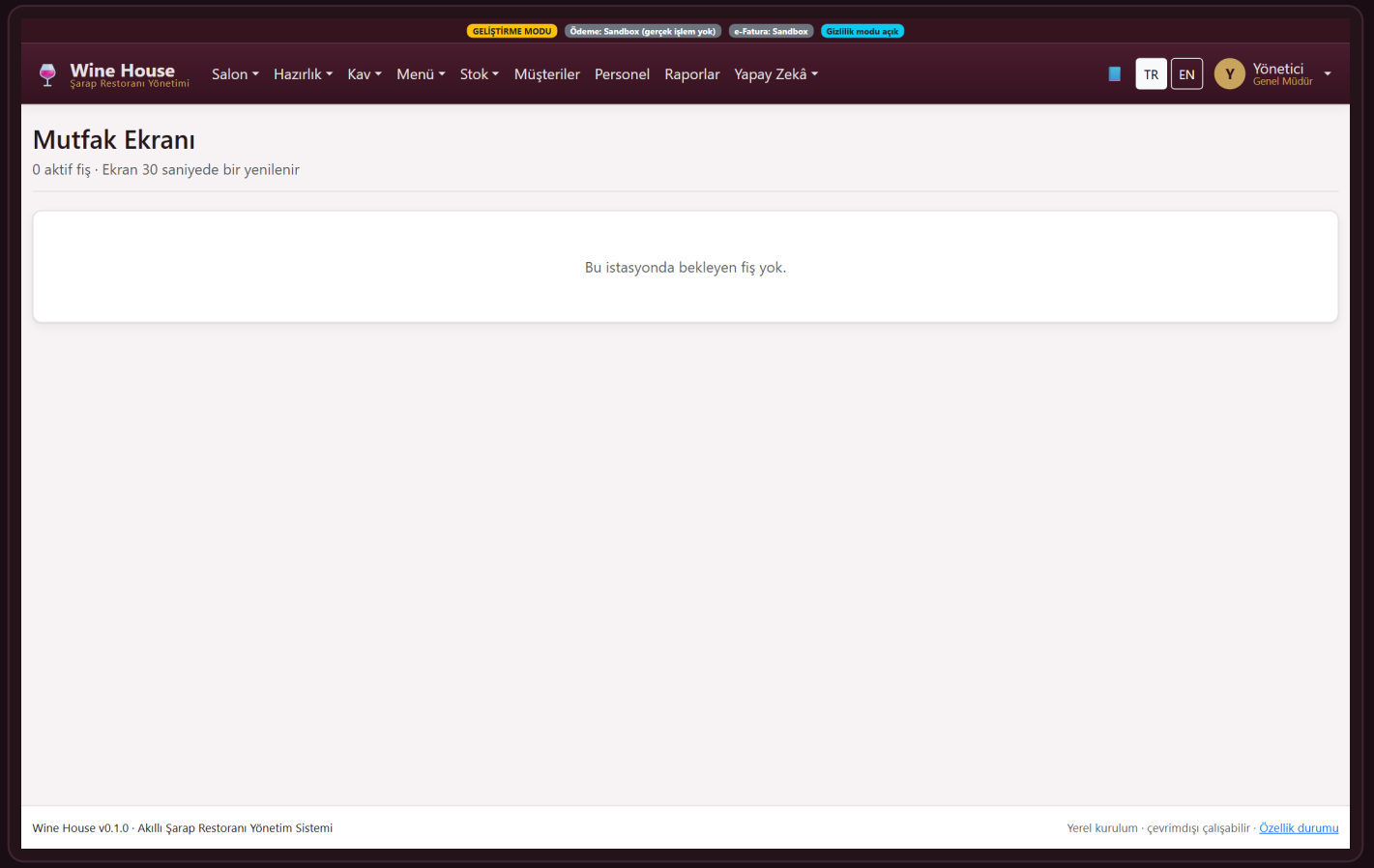

Order management with price freezing, line voids, discounts and approvals

# Reservations



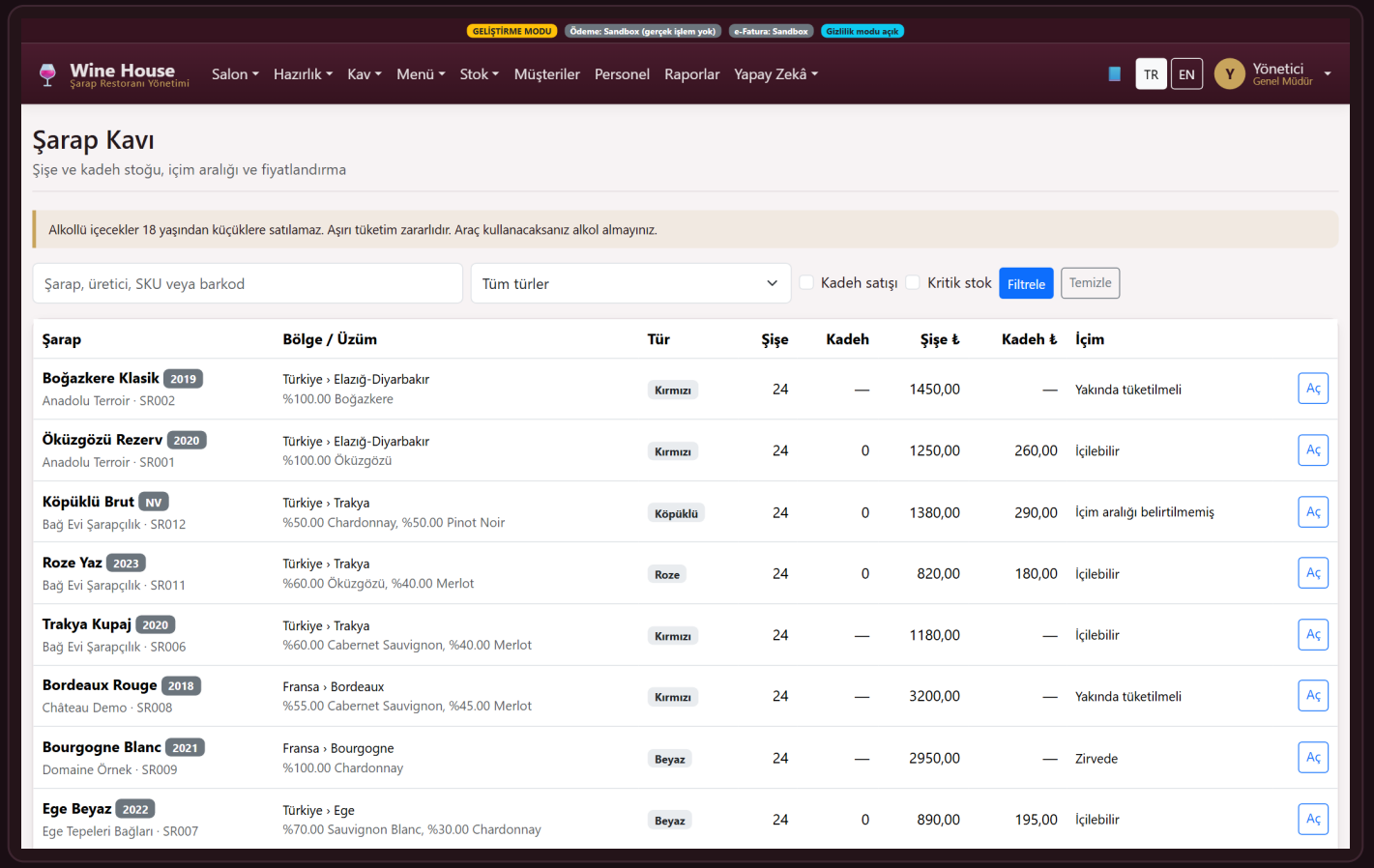

Conflict checking, capacity control and waitlist

# Kitchen display (KDS)



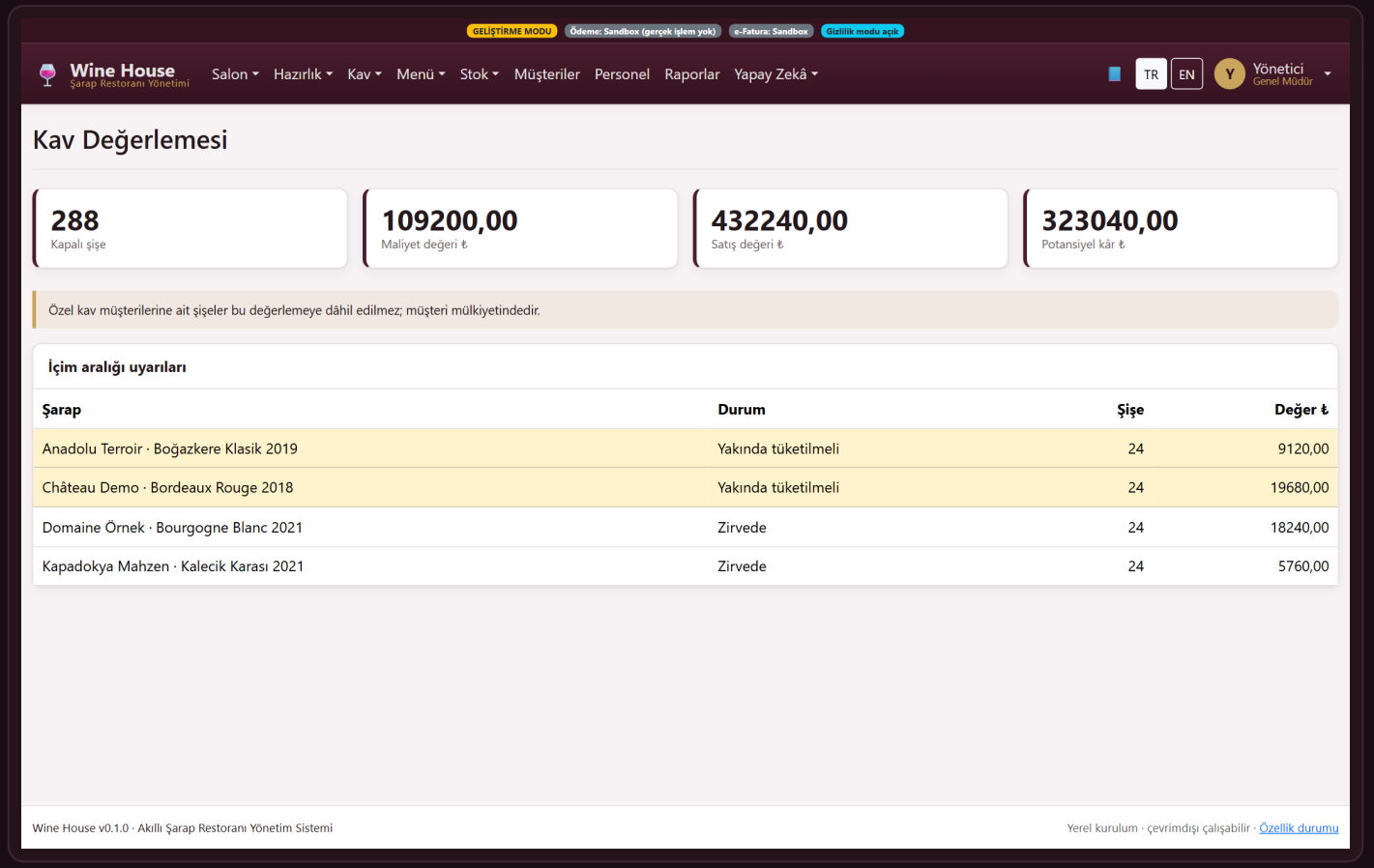

Station separation, ticket progression, delay warnings and allergen flags

# Wine cellar



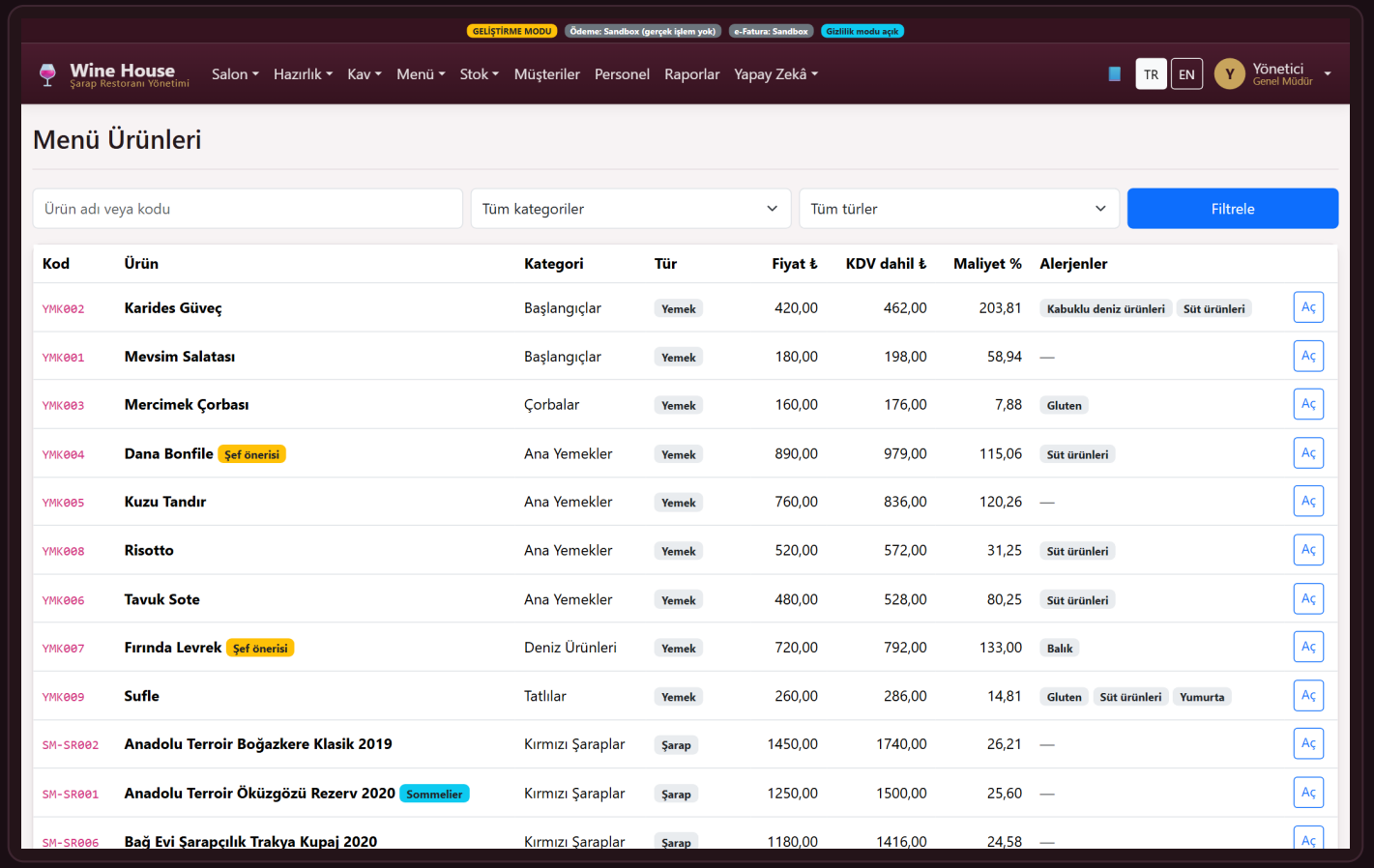

Bottle and by-the-glass stock, drinking windows and pricing

# Cellar valuation



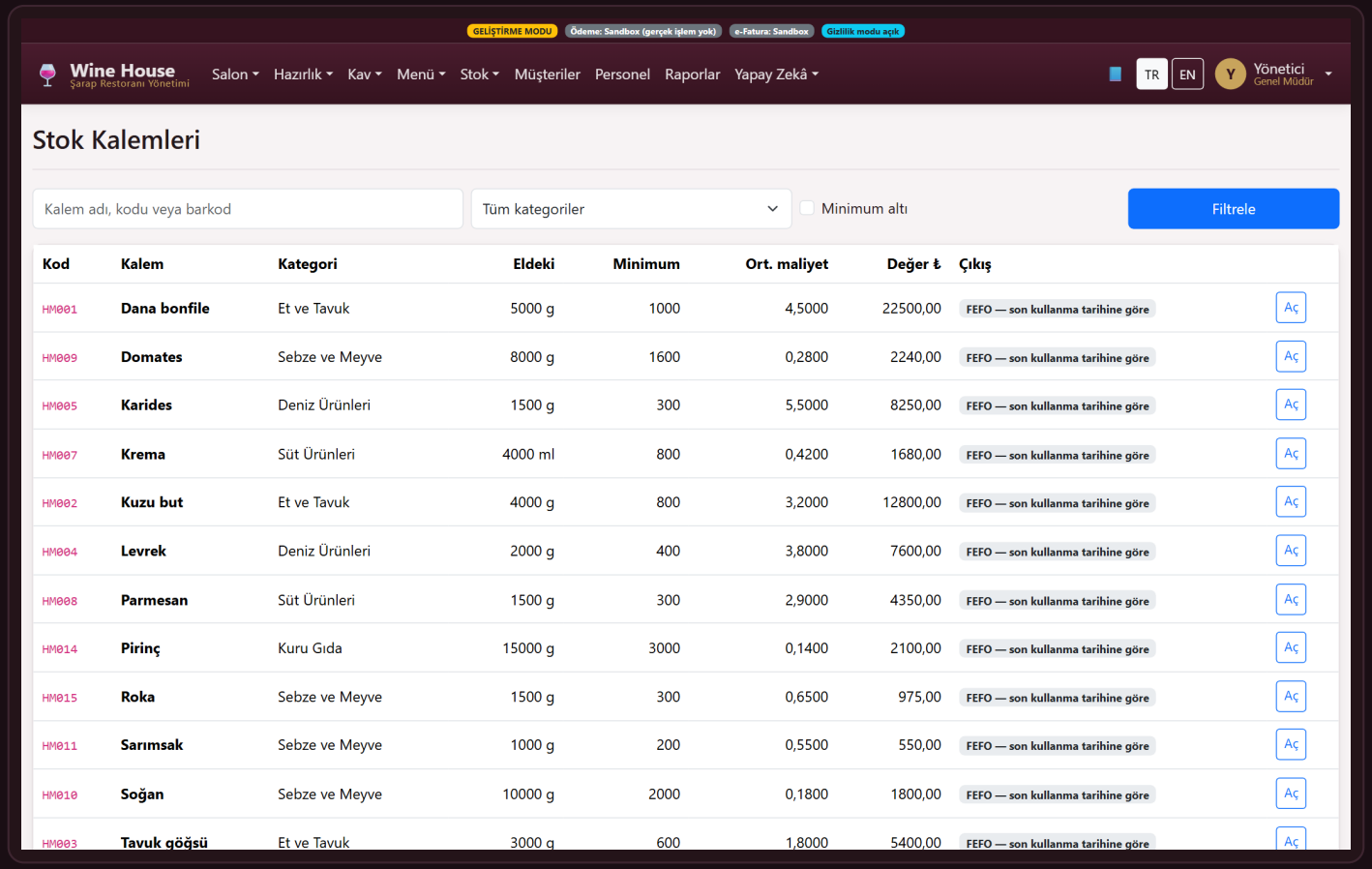

Financial snapshot of the cellar at cost and list value

# Menu and recipes



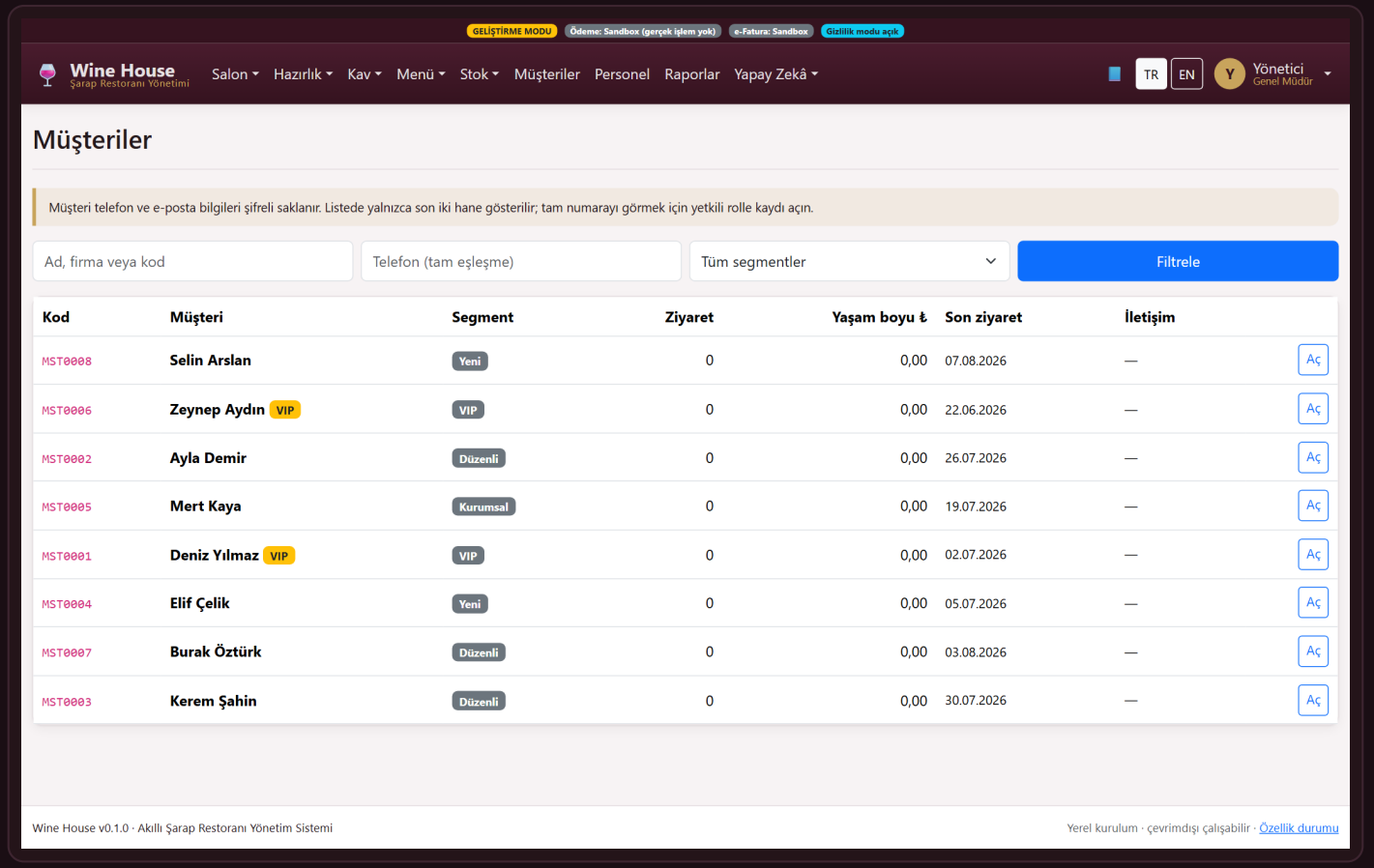

Menu management with portion cost, allergens and nutrition labels

# Inventory and purchasing



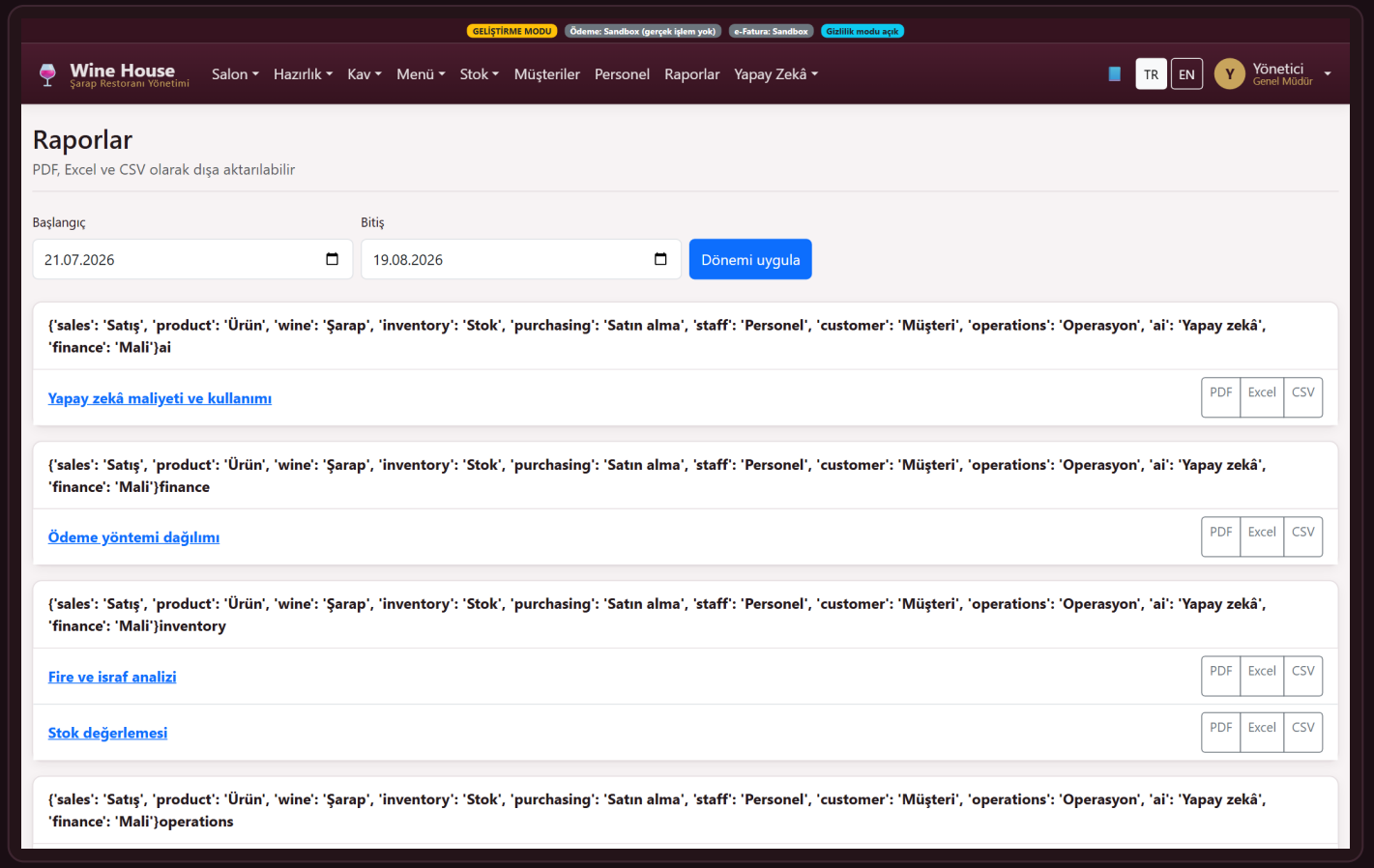

FIFO/FEFO lot tracking, receiving, stock counts and waste flows

# Customer relations



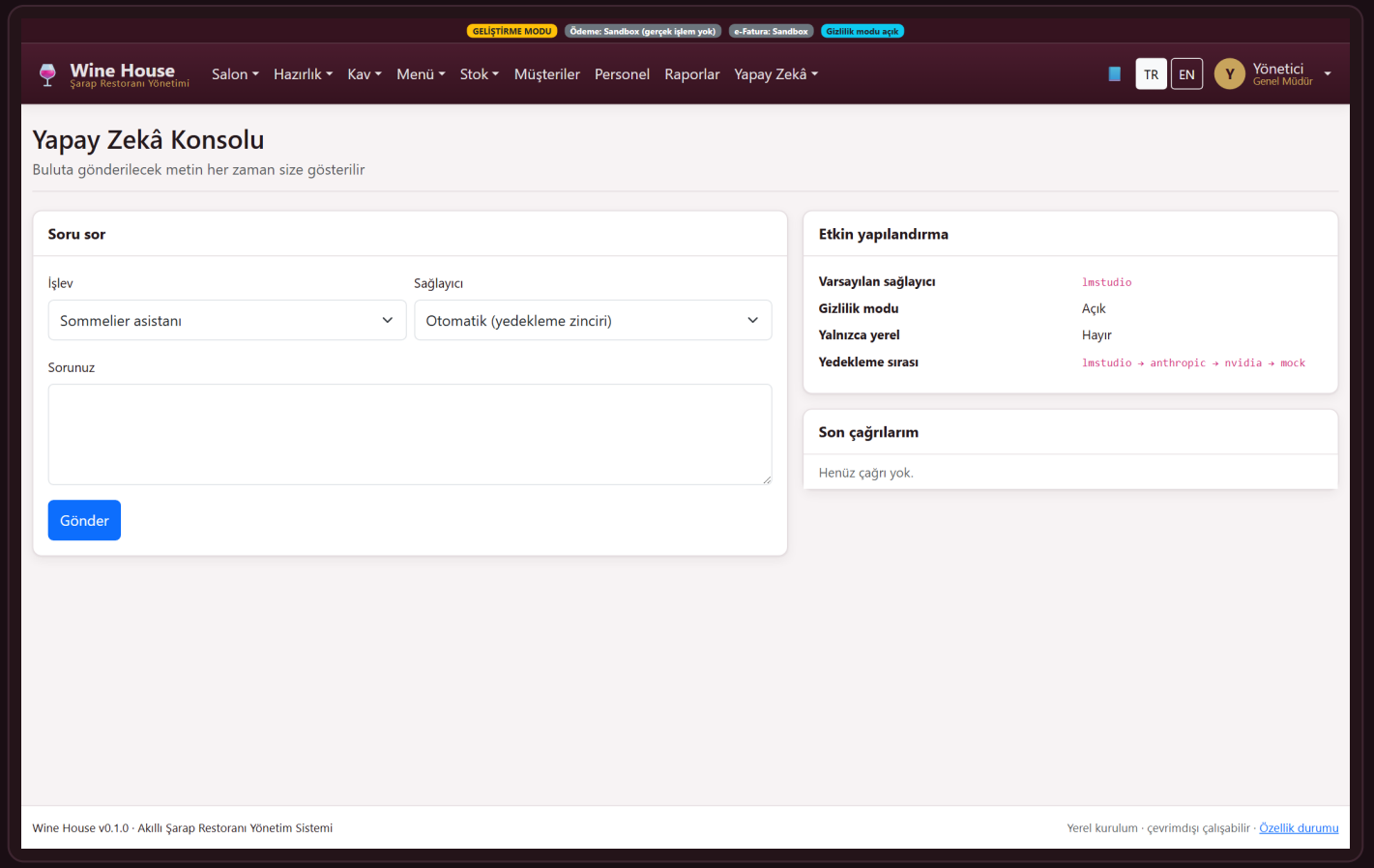

Consent management, export, anonymisation and loyalty — GDPR flows built in

# Reports



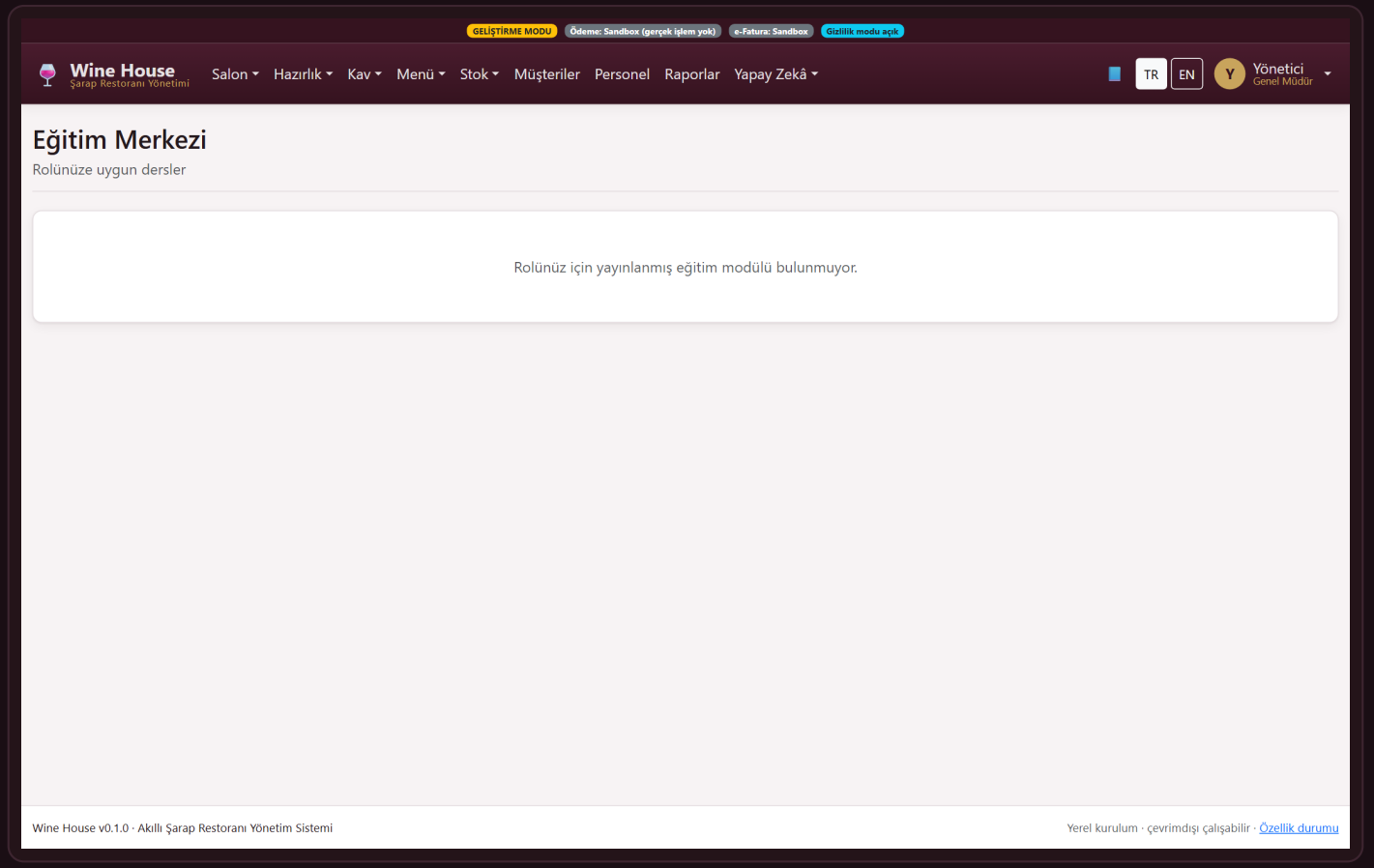

13 ready-made reports · PDF, Excel and CSV export

# AI assistant




Local-first providers, masking and budget control; it advises, never decides

# Training centre




Role-based staff training modules and lessons

# Wine House

## Smart Wine Restaurant Management System

Version 0.1.0 · 2026

- Runs locally · resilient offline
- Freezes financial records and leaves an audit trail
- Deducts by-the-glass sales from stock correctly
- Makes AI an assistant, never an authority
- Encrypts and masks personal data